DATA MANAGEMENT CARPENTRY · EPISODE 5

Choosing Where to Store Your Data

Evaluating local, personal cloud, shared network, and institutional research storage against accessibility, backups, security, and sustainability.

EPISODE SCOPE

Questions & Objectives

Core Questions

Where should research data live?

What are the pros and cons of options?

How to ensure data is backed up & safe?

How to share when publishing research?

Learning Objectives

Differentiate local, network, & cloud tiers.

Evaluate security, backup, and longevity.

Identify personal account departure risks.

Publish citable datasets with DOIs.

CONTEXT

Where Should Data Actually Live?

The Convenience Default

Researchers often begin storing files wherever is immediately handy: a personal laptop, USB drive, or free cloud tier.

This works for an individual initially, but fails as projects expand to new researchers and larger outputs.

The Realities of Research

- Multiple collaborators across institutions
- Multi-gigabyte raw instrument datasets
- Sensitive, confidential, or clinical data
- Long-term funder retention mandates (5–10 years)

REFLECTION EXERCISE

Where Do You Store Your Data?

Spend a few minutes discussing your current storage set-up with a partner:

- Where do you currently store research data?
- Why did you choose that location?
- How do collaborators access it?
- Are the data actively backed up?
- What happens if your laptop dies tomorrow?
- What happens if you suddenly leave the project?

COMMON SOLUTIONS & BLIND SPOTS

Current Storage Landscapes

Where Researchers Keep Data

- Laptop internal drive / USB thumb drives
- Personal OneDrive / Google Drive / Dropbox
- Department shared network drives
- HPC scratch systems / external hard drives

The Unconsidered Blind Spots

Most setups are chosen for personal speed, without considering backup policies, institutional compliance, or transfer of ownership.

A single point of failure can end a research project.

EVALUATION FRAMEWORK

Four Pillars of Good Storage



Accessibility

Can project members and collaborators find and open data?



Recoverability

Can lost or damaged files be reliably restored?



Security

Is sensitive data protected against unauthorised access?



Sustainability

Will the storage and formats remain viable in 5+ years?

EXERCISE & SOLUTION

Recoverability: Features of a Good Backup System

Hardware Fails

Laptops are stolen, SSDs wear out, and accidental deletions occur.

Storage is not backup! Storing a file does not mean it can be restored.

Essential Backup System Traits

- **Automated & Scheduled:** Runs nightly or on connection without manual effort.
- **Off-site Copies:** Stored in a separate building or cloud to survive physical disaster.
- **Accessible Restoration:** Easy for researchers to self-restore past revisions.
- **Redundancy:** Multiple historical snapshots prevent overwriting bad data.

TIER 1: LOCAL STORAGE

Local Storage: Speed vs. Risk

Advantages

- Fastest read/write performance
- Convenient; zero setup time
- Works offline without internet
- Excellent for active daily scratch work

Disadvantages

- Single point of total hardware failure
- Impossible for team members to access
- Physical theft risk (left on trains)
- Fails all four pillars of long-term storage

Takeaway: Great for working caches; disastrous as the sole master home of research data.

TIER 2: PERSONAL CLOUD

Personal Cloud & The Departure Trap

Benefits (OneDrive / Google Drive)

- Multi-device automatic background sync
- Built-in version history and link sharing
- Safer than unencrypted email attachments

The Researcher Departure Trap

A postdoc keeps master project files in personal OneDrive. When they leave the university, IT deactivates the account.

Institutional privacy rules often prohibit administrators from transferring personal account data to colleagues!

TIER 3 & 4: PROJECT & ENTERPRISE

Shared Drives & Research Data Warehouse

Shared Network / SharePoint

- Managed centrally; not tied to an individual
- Granular permissions across project groups
- Requires oversight to prevent bloat

Institutional Storage (RDW)

- Dedicated research infrastructure with automated backups
- High-capacity storage connected directly to HPC clusters
- Survives staff turnover; project-level ownership

For master research data, project-owned institutional storage is the recommended home.

DISCUSSION 1 · INSTRUCTOR WALKTHROUGH

Accessing Different Storage Locations

Follow along in File Explorer / Finder as we inspect:

LOCATIONS INSPECTED

1. Local drive (`C:\` / `Macintosh HD`)
2. Synchronised OneDrive folder
3. Shared network file share
4. Research Data Warehouse (RDW)

EVALUATION CHECKLIST

- Where do path shortcuts lead?
- Who has permissions to edit/delete?
- What are the backup snapshot intervals?
- What happens off-campus?

GOVERNANCE

Permissions & Least Privilege

Role	Permissions	Purpose
Project Lead	Full Control	Manage structure & roles
Active Researchers	Read & Write	Add data, run analysis
External Reviewers	Read Only	Inspect without modification

The Principle of Least Privilege

Grant only the minimum access needed for each role. This protects raw data from accidental deletion or corruption.

Regularly audit permissions to remove access as members complete contracts.

LEGAL COMPLIANCE

Personal & Confidential Data

Special Category Data

Requires enhanced technical and organisational safeguards:

Health data, biometric records, ethnicity, political opinions, religious beliefs.

Must complete a Data Protection Impact Assessment (DPIA).

Legal Mandates Under UK GDPR

- **Subject Access Requests:** Rapid location of records
- **Right to Erasure:** Confident, irreversible deletion
- **Purpose Limitation:** Strict use only for consented goals
- **Breach Notification:** Strict institutional reporting timelines

DATA SECURITY

Encrypting Research Storage

Whole-Disk vs. External Drives

Laptops are often protected with whole-disk encryption (BitLocker / FileVault).

However, **external backup drives are NOT encrypted by default** and must be secured manually!

The Encryption Key Hazard

If you lose your encryption key, your research data is permanently destroyed. There is no back-door recovery.

Note: Windows BitLocker drives may not mount natively on macOS/Linux.

MIGRATION RISKS

Changing Storage Locations

BROKEN HARCODED PATHS

```
Windows:
  D:\ProjectData\
  ↳ R:\Projects\ProjectData\

macOS / Linux:
  /Users/Claire/ProjectData/
  ↳ /data/Projects/ProjectData/
```

When Migrating Master Data:

- Inform all collaborators prior to the move.
- Update documentation and root READMEs.
- Use relative paths in analysis scripts (`./data/raw.csv`) rather than hardcoded system drives.

PUBLISHING RESEARCH

Sharing Outputs via Repositories & DOIs

Open Repositories

Zenodo, Figshare, institutional repositories, or subject-specific archives.

Provides long-term preservation far beyond the lifespan of personal web pages.

Digital Object Identifiers (DOIs)

A persistent identifier providing permanent, stable links to datasets and code.

Enables datasets to be formally cited in academic papers.

DOI RESOLVER EXERCISE

Challenge: Resolving DOIs

Using the DOI resolver (`https://dx.doi.org/`), look up these two identifiers:

1. `10.5281/zenodo.3960218` | 2. `10.1371/journal.pone.0090081`

Identify: Storage repository, object type (software vs. article), creator names, publication date, and citation format.

DOI RESOLVER SOLUTION

Resolved Research Artifacts

DOI: 10.5281/zenodo.3960218

palmerpenguins v0.1.0

Type: Software package with dataset

Platform: Zenodo

Authors: Horst, Presmanes Hill, Gorman (2020)

DOI: 10.1371/journal.pone.0090081

Antarctic Penguin Sexual Dimorphism

Type: Peer-reviewed research article

Platform: PLOS ONE

Authors: Gorman, Williams, Fraser (2014)

PRACTICAL MATCHING EXERCISE

Matching Data to Storage

Where would you store each of these five research assets?

1. Active analysis script files that only you are currently editing
2. A shared collaboration involving five co-investigators
3. Multi-terabyte image dataset processed on an HPC cluster
4. Personally identifiable patient survey responses
5. Final cleaned benchmark dataset accompanying an accepted paper

MATCHING RECOMMENDATIONS

Recommended Storage Allocation

1. **Active solo analysis:** Local working drive with automated institutional cloud backup.

2. **5-person collaboration:** Shared project drive / Institutional SharePoint site.

3. **HPC large dataset:** Institutional Research Data Warehouse (RDW) mounted to HPC nodes.

4. **Personal identifiable data:** Approved secure institutional server with encryption & access control.

5. **Published paper output:** Public repository (Zenodo / Figshare) with a permanent DOI.

SUMMARY

Episode 5 Key Points

Four Pillars: Balance accessibility, recoverability, security, and sustainability.

Avoid Personal Traps: Project-level ownership survives member departures.

Security & GDPR: Encrypt sensitive data; adhere to DPIA protocols.

DOIs for Sharing: Publish final datasets on Zenodo/Figshare for citable impact.

The Invariant: Research data should ideally be stored in project-owned, backed-up institutional systems.

LOOKING AHEAD

Next: Episode 6

We now know where research data should be stored. Next, we explore managing file transfers, storage limits, compression, and reducing data duplication.

Managing Storage and File Transfers